13.13 Finding Perimeter and Area on the Coordinate Plane

Lesson Introduction

Benchmark	MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving the perimeter or area of polygons.		Learning Target	• I can find the perimeter and area of polygons and composite figures drawn on the coordinate plane.
Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
Essential Questions	• How can we find the area of a composite figure? • How can we find the perimeter of a composite figure? • Why is the perimeter of a composite figure not the same as adding the two separate perimeters of the figures it is made of?			
Lesson Narrative	The lesson first has students calculating area and perimeter of two shapes separately, and then introduces them to the idea of composite figures. Students then explore how the area of a composite figure is the same as adding the area of the two figures, while that is not the case with the perimeter.			
Component Summary	13.13.1 Warm-Up (~5 mins)	Writing questions about the Dancing House *(SE p. 355)*	13.13.3 Try It (15 mins)	Practice finding the perimeter and area of composite figures *(SE p. 358)*
	13.13.2 Exploration (25 mins)	Perimeter and area of composite figures *(SE p. 355)*	13.13.4 Wrap-Up (~5 mins)	Reflection *(SE p. 359)*
Resources	Glossary	Materials	Additional Preparation	
	Key Terms: • Composite Figure Additional Glossary: N/A	• Calculators	• N/A	

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may think that to find the area of a parallelogram, it is the same as finding the area of a rectangle, especially since the side lengths could be used for a rectangle. • Students may add the perimeter of each shape of a composite figure, instead of calculating the perimeter of the figure.	• Consider that this lesson is towards the end of the course and that the concepts of finding area and perimeter are not new, therefore, encourage students to use their resources to remember anything that they may have forgotten. This spaced recall can benefit the student by having them struggle to recall the concepts they have previously learned. • In problems that require several steps, students should not round until the final step. Rounding mid-step could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values.
	Literacy and Language Routines	**Mathematical Thinking and Reasoning (MTR) Standard**
	Compare and Connect In this lesson, students are identifying and comparing their approach to finding side lengths with that of other students in the class, while also determining which method is best. Students are also making connections to finding the perimeter and area of individual shapes, and how that can help us find the area and perimeter of composite figures.	**MA.K12.MTR.1.1: Participate in Effortful Learning** Students are analyzing problems in a way that makes sense for them given the task, such as when finding the area and perimeter of a composite figure. While it is hoped that students see the pattern to find the area and perimeter, students are still encouraged to analyze the problem on their own. This allows students to stay positive and engaged throughout the lesson.
Differentiate Instruction	Consider providing anchor charts, or reviewing the various area formulas and what each variable represents, so that students have that extra support in this lesson. Also, consider providing students with ways to find the length of diagonal lines, and consider performing an example so that students have an example to go off of when calculating the side lengths of shapes in this lesson. Additional differentiation opportunities are denoted throughout the lesson guide.	

Lesson Notes:

13.13.1 Warm-Up: The Dancing House

Component Overview: *Students are given an opportunity to observe a photo and create two questions that come to their minds. One question is what first comes to their minds, and the other question is one they know a 5th grader could answer.*

13.13.1 Warm-Up: The Dancing House

The Dancing House is located in Prague, Czech Republic, and was designed in 1992. The style of this building is known as deconstructivist architecture due to its unusual shape.

1. What is the first question that comes to your mind?
   Answers will vary. Is the structure of the twisted building as stable as a building that is not twisted?

2. Write a question that a 5th grader could answer using the photo.
   Answers will vary. What is an appropriate estimation for the number of windows in the twisted structure?

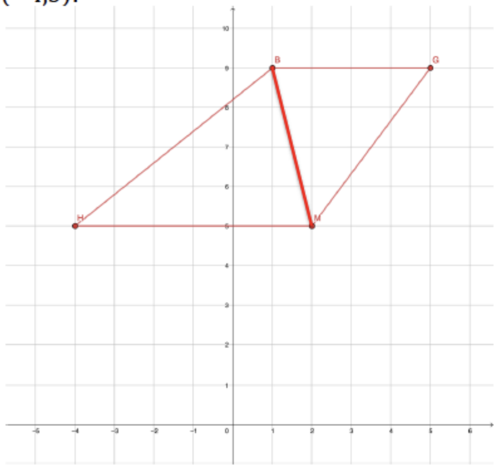
Consider an opportunity to have a discussion with students about how they created the question for questions. The discussion can include how they knew what topics to ask in their questions, such as which topics were too easy for a 5th grader versus which questions would be too difficult.

13.13.2 Exploration: Perimeter and Area

Component Overview: *In this section, students are finding the area and perimeter of a trapezoid and parallelogram separately, and then they are given a composite shape that is created from the same parallelogram and trapezoid and are asked to find the perimeter and area of the shape. The purpose of this section is to introduce students to composite figures, and to notice that the area of a composite figure is the same as adding the area of the two shapes, while the perimeter is not calculated the same way.*

13.13.2 Exploration: Perimeter and Area

1. Trapezoid $BGMH$ has vertices at $B(1,9)$, $G(5,9)$, $M(2,5)$, and $H(-4,5)$.



   a. Graph $BGMH$.
   Image found on graph.

   b. Complete the table.

Line Segment	Length
$\overline{BG}$	$BG = 4$
$\overline{GM}$	$GM = 5$
$\overline{MH}$	$MH = 6$
$\overline{HB}$	$HB = \sqrt{41}$

This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. Students utilize skills and concepts learned previously in the course. Since this is content that is spiraled, allow students to utilize their own resources to recall formulas. Monitor while students are working to determine if there are any questions that need to be debriefed before 13.3.3 Try It. Students may revise their thinking throughout the Exploration as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group before releasing students to complete 13.3.3 Try It.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

c. What is the perimeter of $BGMH$?
21.403 units

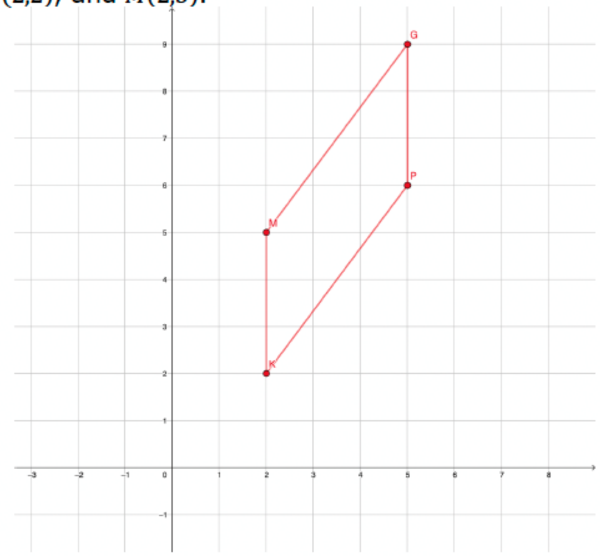
d. Draw a diagonal line from B to M to make ΔBMH and ΔBGM.
Image found on graph.

e. Find the area of ΔBMH and ΔBGM.

$\Delta BMH =$ ____*12 units²*____ $\Delta BGM =$ ____*8 units²*____

f. What is the area of $BGMH$?
20 *square units*

2. Parallelogram $GPKM$ has vertices at $G(5,9)$, $P(5,6)$, $K(2,2)$, and $M(2,5)$.



a. Graph $GPKM$.
Image found on graph.

- *Why do you think it is needed to separate BGMH into two separate triangles?*
- *If $BGMH$ was a rectangle, would separating it into two separate triangles to find the area be the same as using the equation to find the area of a rectangle?*
- *What other quadrilaterals do you think it will be beneficial to separate into two triangles in order to find the area?*

13.13.2 Exploration: Perimeter and Area (continued)

From your observations of students on question 1, consider whether some students could benefit from a check-in with you, or if utilizing a small group to work through question 2 with you could be beneficial to students who struggled with question 1.

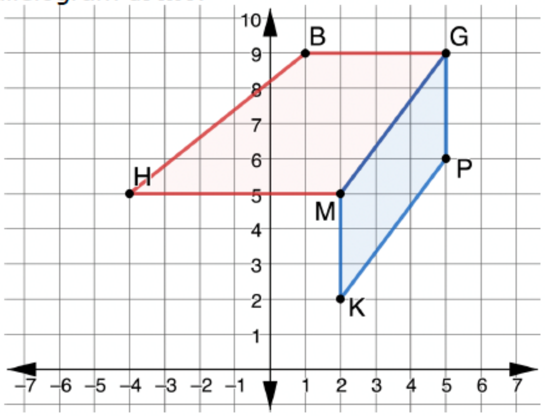
b. Complete the table.

Line Segment	Length
$\overline{GP}$	3
$\overline{PK}$	5
$\overline{KM}$	3
$\overline{MG}$	5

c. What is the perimeter of $GPKM$?
16 *units*

d. What is the area of $GPKM$?
9 *squared units*

3. The figure shown is created by trapezoid $BGMH$ and parallelogram $GPKM$.



a. What do you notice about this figure?
Answers will vary. Trapezoid $BGMH$ and parallelogram $GPKM$ share the side $\overline{GM}$

A ***composite figure*** is a two- or three-dimensional figure that can be decomposed into smaller figures.

b. Explain why this figure is considered a composite figure.
Answers will vary. The figure is composed of trapezoid $BGMH$ and parallelogram $GPKM$ and can be broken up into the two shapes.

c. How could you find the perimeter of a composite figure?
Answers will vary. To find the perimeter of a composite figure, add up all of the side lengths that are on the perimeter of the figure, but do not include any sides they share like $\overline{GM}$ in the figure above.

d. What is the perimeter of composite figure $BGPKMH$?
27.403 *units*

e. Explain why the perimeter of the composite figure is not the sum of the perimeters of the parallelogram and the trapezoid.
Answers will vary. The perimeter of trapezoid $BGMH$ and parallelogram $GPKM$ both include the side length $\overline{GM}$ which is no longer considered an edge of the composite figure.

f. What is the area of $BGPKMH$?
29 *square units*

- *Why can't we find the area of parallelogram GPKM using the equation, $b \cdot h$?*
- *Why isn't the area of a parallelogram and rectangle the same?*
- *Why does it not produce the same result?*

Consider an opportunity to take all of the student responses and create a Gallery Walk or class discussion so that students can share what they notice with others. If turning this into an activity, consider having students write down 1 – 2 things that other students noticed that they did not.

Could we create a composite figure with trapezoid $BGMH$ and parallelogram $GPKM$ that would have the same perimeter as adding up the perimeters of trapezoid $BGMH$ and parallelogram $GPKM$?

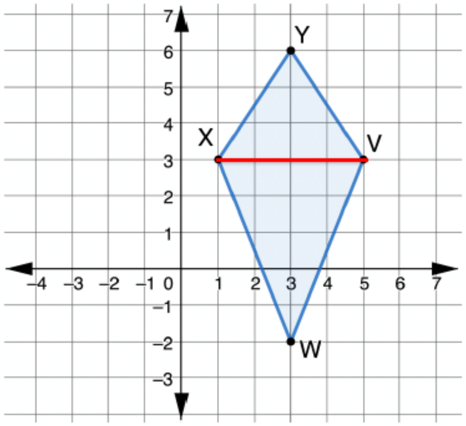
13.13.3 Try It: Perimeter and Area

Component Overview: *Students are extending the idea from the Exploration to another problem involving composite figures.*

13.13.3 Try It: Perimeter and Area

Use observational data from the 13.13.2 Exploration to determine how much guidance is needed to bring together key ideas before releasing students to complete this component. If necessary, consider modeling Question 1 to address common misconceptions shown in the Exploration.

1. Kite $YVWX$ is shown with vertices at $Y(3,6)$, $V(5,3)$, $W(3,-2)$, and $X(1,3)$.



a. Complete the table.

Line Segment	Length
$\overline{YV}$	$YV = \sqrt{13} \approx 3.606$
$\overline{VW}$	$VW = \sqrt{29} \approx 5.385$
$\overline{WX}$	$WX = \sqrt{29} \approx 5.385$
$\overline{XY}$	$XY = \sqrt{13} \approx 3.606$

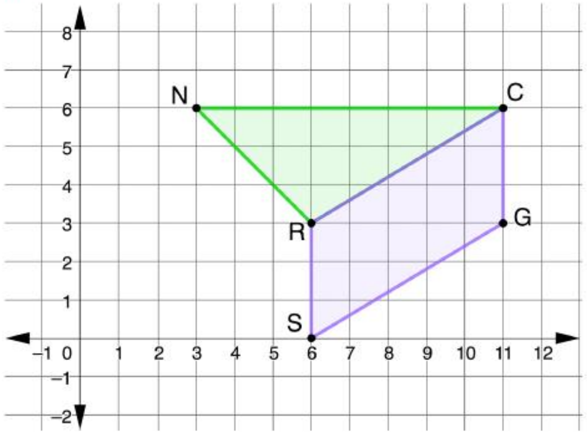
b. What is the perimeter of $YVWX$?
17.981 *units*

c. Draw line XV to create ΔYVX and ΔVWX.
Image found on graph.

d. Find the area of each triangle.
$\Delta YVX =$ ___6 *units*^{2}___ $\Delta VWX =$ ___10 *units*^{2}___

e. What is the area of $YVWX$?
16 *square units*

2. The composite figure shown is created by ΔNCR and parallelogram $RCGS$. $N(3,6)$, $C(11,6)$, $G(11,3)$, $S(6,0)$, and $R(6,3)$.



a. Complete the table. Round your answer to the nearest thousandth.

Figure	Perimeter	Area
ΔNCR	18.074 *units*	12 *square units*
$RCGS$	17.662 *units*	15 *square units*

b. What is the perimeter of the composite figure?
24.074 *units*

c. What is the area of the composite figure?
27 *square units*

- *Why is dividing Kite $YVWX$ into two triangles helpful?*
- *What other ways could Kite $YVWX$ be divided?*
- *What is the perimeter of triangle YVX and VWX?*
- *Why is the perimeter of the kite YVWX not the same as adding the perimeter of triangle YVX and VWX?*

If working with students who are struggling, consider providing students a few minutes to write down things they notice and wonder about the composite figure. This will allow students to make observations before finding the perimeter and area.

For students who finish early, challenge them to create a third shape on the composite figure and find the area and perimeter of the additional shape (alone), and then the area and perimeter of the new composite function.

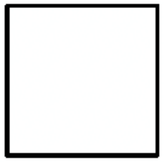
13.13.4 Wrap-Up: Cool Down

Component Overview: *Students complete a Cool Down that has a theme with shapes. Consider an opportunity to use this to collect data about where students are struggling and what they understand.*

13.13.4 Wrap-Up: Cool Down

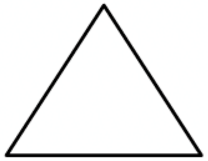
Consider creating a rubric with ideas that you would expect students to have written down in these 3 questions, that you would consider them to have mastered and understood this lesson.

1. What is something that is now "squared away" in your mind about finding the perimeter and area of polygons and composite figures drawn on a coordinate plane?



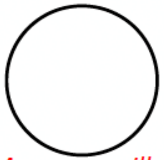
Answers will vary.

2. What are three "points" that are important about finding the perimeter and area of polygons and composite figures drawn on a coordinate plane?



Answers will vary.

3. What is something that is still "circling" your brain about finding the perimeter and area of polygons and composite figures drawn on a coordinate plane?



Answers will vary.